

Lizhe Zhang

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Research Interests: Agent Safety & Security • LLM Evaluation • Trustworthy AI

EDUCATION

University of California, Santa Cruz , Santa Cruz, CA <i>Ph.D. in Computer Science and Engineering</i>	Sep. 2026 – Present Advisor: Prof. Chenguang Wang
University of California, San Diego , La Jolla, CA <i>M.S. in Computer Science</i>	Sep. 2024 – Mar. 2026 GPA: 4.00/4.00
Emory University & Georgia Institute of Technology , Atlanta, GA <i>B.S. in Physics, Minor in Applied Mathematics (Emory)</i> <i>B.S. in Computer Engineering (Georgia Tech)</i>	Aug. 2019 – Dec. 2023 GPA: 3.90/4.00 GPA: 3.96/4.00

RESEARCH EXPERIENCE

University of California, Santa Cruz <i>Graduate Student Researcher</i>	Sep. 2026 – Present Advisor: Prof. Chenguang Wang
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- Developing evaluations for malicious behavior and unsafe tool use in LLM-based agents.
- Studying how autonomous agents plan and execute multi-step harmful actions across external tools.

University of California, San Diego <i>Graduate Researcher</i>	Jun. 2024 – Mar. 2026 Advisor: Prof. Jingbo Shang
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- Developed a code-rewriting framework to diagnose harmful memorization in LLM code generation.
- Introduced the **Harmful Memorization Index (HMI)** to distinguish brittle reuse from benign code reuse.
- Evaluated Qwen and Llama models on MBPP+ and BigCodeBench, analyzing scaling and SFT/PPO effects on HMI.

Rice University <i>Graduate Researcher</i>	Sep. 2024 – May 2025 Advisor: Prof. Xia Hu
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- Developed an uncertainty-based SLM-to-LLM routing framework balancing accuracy, latency, and inference cost.
- Benchmarked 8 small LMs, 2 large LMs, and 8 uncertainty-estimation methods across multiple tasks.
- Developed a data-agnostic proxy routing strategy that generalizes to unseen datasets without per-dataset tuning.

Georgia Institute of Technology <i>Undergraduate Researcher</i>	May 2023 – Feb. 2024 Advisor: Prof. Thomas Plötz
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- Implemented and ran virtual-IMU generation pipelines for IMUGPT 2.0 and wearable emotion recognition.
- Conducted large-scale experiments across motion datasets and prepared synthetic sensor data for model evaluation.

PUBLICATIONS

Evaluating Harmful Memorization in LLM Code Generation <i>Lizhe Zhang*</i> , Wentao Chen*, Li Zhong, Letian Peng, Zilong Wang, Jingbo Shang	EMNLP 2026 Findings [paper]
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Confident or Seek Stronger: Exploring Uncertainty-Based On-device LLM Routing From Benchmarking to Generalization <i>Yu-Neng Chuang*</i> , Leisheng Yu*, Guanchu Wang, Lizhe Zhang , Zirui Liu, Xuanning Cai, Yang Sui, Vladimir Braverman, Xia Hu	NeurIPS 2025 Workshop [paper]
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IMUGPT 2.0: Language-Based Cross Modality Transfer for Sensor-Based Human Activity Recognition <i>Zikang Leng, Amitrajit Bhattacharjee, Hrudhai Rajasekhar, Lizhe Zhang, Elizabeth Bruda, Hyeokhyen Kwon, Thomas Plötz</i>	IMWUT 2024 [paper]
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Emotion Recognition on the Go: Utilizing Wearable IMUs for Personalized Emotion Recognition <i>Zikang Leng, Myeongul Jung, Sungjin Hwang, Seungwoo Oh, Lizhe Zhang, Thomas Plötz, Kwanguk Kim</i>	UbiComp/ISWC 2024 [paper]
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ACADEMIC SERVICE

Organizer , Agents in the Wild: Safety, Security, and Beyond	Proposed for ICLR 2027
Reviewer , Conference on Language Modeling (COLM)	2026

WORK EXPERIENCE

Machine Learning Intern

Horizon Robotics

May 2023 – Aug. 2023

Beijing, China

- Benchmarked state-of-the-art 3D semantic segmentation models on SemanticKITTI for autonomous-driving perception.
- Identified failure modes of standard mIoU when evaluating small spatial deviations in 3D predictions.
- Prototyped a range-aware evaluation framework with the perception team for more informative model comparison.

Software Engineering Intern

Inspur Group

Jun. 2021 – Aug. 2021

Jinan, China

- Replaced a Haar-cascade detector with YOLOv5, improving license-plate detection accuracy by over 40%.

SELECTED PROJECT

Find Your Spot

Georgia Institute of Technology, Senior Design Project

Aug. 2023 – Dec. 2023

[\[poster\]](#)

- Built an end-to-end indoor parking navigation system using BLE beacons and RSSI-based localization to guide drivers to available parking spaces.
- Developed the backend for processing IoT sensor and beacon data, integrating localization and routing algorithms for real-time parking navigation.

PATENTS & HONORS

Language-based Cross-modality Transfer System for Generating Virtual Inertial Measurement Unit Data to Enhance Human Activity Recognition

U.S. Patent Application No. 19/566,099

Filed Mar. 2026

Sigma Pi Sigma National Physics Honor Society

Emory University, 2023

Faculty Honors

Georgia Institute of Technology, 2022–2023

TEACHING EXPERIENCE

Teaching Assistant, CS 170: Introduction to Computer Science, Emory University

Fall 2021, Spring 2022

SKILLS

Programming: Python, C/C++, Java, Swift

Machine Learning: PyTorch, Hugging Face Transformers, TensorFlow, scikit-learn, OpenCV

Tools: Git, Docker, vLLM, LaTeX